

2. [Maximum mark: 8]

The time, T minutes, that customers wait in line at a coffee shop is normally distributed with a mean of 4.5 minutes and a standard deviation of 1.2 minutes.

(a) Find $P(T > 6)$. [2]

(b) Find $P(3 < T < 6)$. [2]

(c) On a particular morning, 10 customers each wait in line independently. Find the probability that exactly 2 of them wait more than 6 minutes. [2]

(d) A customer is selected at random. Given that they waited more than 3 minutes, find the probability that they waited more than 6 minutes. [2]